1 Notation and conventions

Indices and sets. We use lower-case Latin letters for indices and calligraphic upper-case for sets:

- $r \in \mathcal{R}$ is the receiver. Single-receiver modes (PPP, PPP-RTK) have $|\mathcal{R}| = 1$; baseline modes have a fixed reference and one or more rovers; the network producer (`PPP_RTK_Network`) carries a peer set of $|\mathcal{R}| \geq 2$ stations (50–305 in practice) with no reference/rover asymmetry.
- $g \in \mathcal{G} = \{\text{GPS}, \text{GAL}, \text{BDS}, \text{GLO}, \text{QZS}, \dots\}$ is the constellation.
- $i \in \mathcal{S}_g$ is a satellite of constellation g. p_g denotes the pivot satellite of constellation g under the BSSD parameterisation (Section 4).
- $s \in \mathcal{B}$ is the signal/observation type, identified by its RINEX-3 three-character code (e.g. `C1C`, `L2W`, `L5X`).
- $k \in \mathbb{N}$ is the discrete epoch index.

Vectors and matrices. Bold lower-case for vectors ($\boldsymbol{x}$, $\boldsymbol{r}$); bold upper-case for matrices ($\boldsymbol{H}$, $\boldsymbol{P}$). $\boldsymbol{a}^{\mathrm{T}}$ is the transpose. A hat denotes a state estimate; a tilde a one-step prediction: $\hat{\boldsymbol{x}}_{k|k}$ vs. $\tilde{\boldsymbol{x}}_{k|k-1}$.

Observables. For each receiver/satellite/signal triple, the engine consumes two real-valued observables (in metres) per epoch:

- $P^i_{r,s}$ is the pseudorange (code), in metres.
- $\Phi^i_{r,s}$ is the carrier phase, scaled to metres via λ_s. Beyond the range, clock, and atmosphere terms it shares with the code, its ambiguous (non-range) part is $\lambda_s N^i_{r,s}$ plus the fractional phase biases $b_{r,s} - b^i_s$ (Equation (5)).

Where the distinction matters, s may further denote the band ($L_1, L_2, L_5, \dots$) and the tracking attribute (e.g. `C`, `W`, `X`). The leading observable-type letter (`C`/`L`) selects code vs. phase; the frequency-only quantities f_s, λ_s, and $N^i_{r,s}$ depend only on the (band, attribute) part, so a code and its sibling phase share them. This matches the state keys, which carry the full phase code for ambiguities (`"G01_L1C"`) but only the band digit for receiver code biases (`"GPS_IFB2"`).

Frequencies and wavelengths. f_s is the carrier frequency (Hz) and $\lambda_s = c/f_s$ the wavelength (m). The frequency-squared scaling factor used to project the L1-equivalent slant ionospheric delay onto signal s is

$$\mu_s \;:=\; \frac{f_1^2}{f_s^2}. \tag{1}$$

Here f_1 is the GPS L1 frequency (1575.42 MHz), used as the common reference for *every* constellation, not the band-1 frequency of the satellite's own system. The estimated state $I^i_{r,1}$ is therefore GPS-L1-equivalent even for BDS/GLO/GAL satellites (`MeasurementModel::buildJacobianRow`: `f_ref = Constants::GPS_L1_FREQ`).

Differencing operators. Three differencing operations recur throughout the document:

$$\Delta q^{i,p_g}_{r,s} := q^i_{r,s} - q^{p_g}_{r,s}, \quad \nabla q^i_{r_1 r_2,s} := q^i_{r_1,s} - q^i_{r_2,s}, \quad \nabla\Delta q^{i,p_g}_{r_1 r_2,s} := \Delta\big(\nabla q^i_{r_1 r_2,s}\big).$$

Read Δq as "between-satellite single difference of q relative to pivot p_g," ∇q as "between-receiver single difference," and $\nabla\Delta q$ as "double difference."

Sign conventions. We adopt the convention that the slant ionospheric delay $I^i_{r,s}$ enters *additively* on the code observable and *subtractively* on the carrier phase observable; this is the source of the well-known "code–phase divergence." All other range-additive terms (geometry, clocks, troposphere, biases) carry the same sign on both observables. See Equations (4) and (5) for the explicit equations.

Symbol table.

Symbol	Units	Meaning
$\boldsymbol{r}_r, \boldsymbol{r}^i$	m	ECEF position of receiver/satellite
ρ^i_r	m	geometric range $\|\boldsymbol{r}^i - \boldsymbol{r}_r\|$
$\boldsymbol{e}^i_r$	none	unit LOS from r to i in ECEF
$c\,\mathrm{d}t_r$	m	receiver clock offset (scaled by c)
$c\,\mathrm{d}t^i$	m	satellite clock offset (CLK product in user modes; estimated state `"<sat>|SC"` in `PPP_RTK_Network`)
$c\,\delta t^{(g)}_r$	m	receiver inter-system time offset for constellation $g \neq g_{\mathrm{ref}}$
T^i_r	m	total tropospheric path delay
$\mathrm{ZHD}^i_r, \mathrm{ZWD}^i_r$	m	zenith hydrostatic / wet components
$I^i_{r,1}$	m	L1-equivalent slant ionospheric delay
$N^i_{r,s}$	cyc	integer carrier-phase ambiguity
$b_{r,s}, b^i_s$	m	receiver / satellite phase biases
$\bar{b}^i_s$	m	estimable satellite phase bias (network mode; Section 2)
$B_{r,s}, B^i_s$	m	receiver / satellite code biases (OSB)
$B^{(g,b)}_{r,k}$	m	per-FCN-slot GLONASS receiver IFB, band b; here $k \in [-7, +6]$ is the signed FDMA channel number (a context-local reuse, not the epoch index)
$\varepsilon^P, \varepsilon^\Phi$	m	residual (multipath + thermal)
$\boldsymbol{v}_r$	m/s	receiver velocity (optional / IMU state)
$\lambda_{\mathrm{WL}},\ \lambda_{\mathrm{NL}},\ \lambda_{\mathrm{EWL}}$	m	wide- / narrow- / extra-wide-lane wavelength $c/(f_a \mp f_b)$ for the lane's band pair
λ_{IF}	m	effective ionosphere-free wavelength $\|c_1\lambda_1 - c_2\lambda_2\|$ with the IF coefficients $c_1 - c_2 = 1$; not an integer wavelength (the cycle-slip document)
$\boldsymbol{C}^e_b$	none	body-to-ECEF attitude (direction-cosine) matrix

2 State vector layout

The Kalman filter operates on a state vector $\boldsymbol{x}_k \in \mathbb{R}^{n_k}$ whose dimension n_k varies across epochs. A per-satellite or per-(satellite, signal) state is appended when its satellite or signal is first required, and removed when it is absent from the epoch's measurement set (for example, after the satellite sets below the elevation mask). A cycle slip does not remove the ambiguity state; it re-seeds it (see the ambiguity paragraph below). `StateManager` holds all bookkeeping: the key→index maps, the appending of rows, and the compaction of $\boldsymbol{x}$ and $\boldsymbol{P}$ on removal, which shifts every later index down by one (there is no free list). `KalmanFilter` reads the state through that interface and never indexes it by hard-coded offsets.

2.1 Schematic layout

In schematic form, suppressing the epoch index k:

$$\begin{aligned}\boldsymbol{x} \;=\; \big[\; & \underbrace{\boldsymbol{r}_r^{\mathrm{T}}}_{3},\ \underbrace{\boldsymbol{v}_r^{\mathrm{T}}}_{0\text{ or }3},\ \underbrace{c\,\mathrm{d}t_r}_{1},\ \underbrace{(c\,\delta t_r^{(g)})^{\mathrm{T}}_{g\neq g_{\mathrm{ref}}}}_{|\mathcal{G}|-1},\\ & \underbrace{\mathrm{ZWD}_r,\ (g_N, g_E)}_{1+0\text{ or }1+2},\ \underbrace{(c\,\mathrm{d}t_r^{\phi})}_{0\text{ or }1},\\ & \underbrace{(I_{r,1}^{i})^{\mathrm{T}}_{i\in\mathcal{V}_k}}_{|\mathcal{V}_k|},\ \underbrace{(N_{r,s}^{i})^{\mathrm{T}}_{(i,s)\in\mathcal{A}_k}}_{|\mathcal{A}_k|},\\ & \underbrace{(B_{r,s})^{\mathrm{T}}_{s\in\mathcal{B}}}_{|\mathcal{B}|},\ \underbrace{(B_s^{i})^{\mathrm{T}}_{(i,s)\in\mathcal{D}_k}}_{|\mathcal{D}_k|}\;\big]^{\mathrm{T}}. \end{aligned} \tag{2}$$

where

- $\mathcal{V}_k$ is the set of satellites visible at epoch k (above the elevation mask, with valid orbit and clock products and at least one tracked signal);
- $\mathcal{A}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the set of phase-tracked (satellite, signal) pairs;
- $\mathcal{D}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the subset of (satellite, signal) pairs that carry a satellite code-bias state: the pairs for which the BIA product lacks a satellite OSB or, in the network producer's satellite-OSB mode, every raw code pair (see `estimation.estimate_sat_biases` and `network.estimate_sat_code_bias`).

The schematic omits several conditional refinements, detailed below: the rate states that accompany velocity (a receiver clock rate and one inter-system clock-rate offset per non-reference constellation, both absent under BSSD); the optional per-satellite slant-ionosphere rate under `enable_iono_rate`; the receiver inter-signal code biases of the network producer; the network-producer satellite-clock, satellite-phase-bias and receiver-phase-bias states; the nine-state inertial augmentation of the inertial-aiding document; and the removal of the whole receiver-side clock and inter-system-offset block under the BSSD parameterisation (Section 4.1).

The schematic groups the states by physical role; it is not the storage order. `StateManager` places a fixed core first, in the order position, receiver clock (absent under BSSD), ZWD, velocity, receiver clock rate, phase clock, the two troposphere gradients, the nine inertial states, the inter-system time offsets and the inter-system clock-rate offsets, each present only when enabled. Every dynamic state (ionosphere, ionosphere rate, ambiguity, receiver code bias, satellite code bias, satellite phase bias, receiver phase bias, satellite clock) is appended after the core in order of creation, so the categories interleave; each category is addressed only through its key map, as described under dynamic indexing below.

2.2 Component-by-component description

Position $\boldsymbol{r}_r \in \mathbb{R}^3$. ECEF Cartesian position of the antenna reference point. Always present. `Application::initializePosition` seeds it from the RINEX header approximate position, shifted from the marker to the antenna reference point by the header's antenna H/E/N offset (or from the Station Information File coordinate when that a-priori source is enabled). The seed must lie within the linearisation radius of the geometric range model (Section 3.3). The per-epoch Weighted Least-Squares (WLS) solution seeds the receiver clock (on the first epoch, after a clock jump and

after a data gap); it re-seeds the position only in the kinematic gap-recovery branch of `Application::handleGap`.

Velocity $\boldsymbol{v}_r \in \mathbb{R}^3$. Optional. Enabled by `estimation.estimate_velocity = true`; tightly-coupled IMU integration forces it on. With velocity present, position and velocity form a coupled continuous-white-noise-acceleration (CWNA, integrated-random-walk) block driven by a single acceleration spectral density, `process_noise.acceleration_psd_q` (the CWNA process-noise model in the estimation document); the scalar `position_q` of the velocity-free case is then unused. When BSSD is off, velocity estimation also adds a receiver clock-rate state $c\,\dot{\mathrm{d}}t_r$ and one inter-system clock-rate offset (ISBR) per non-reference constellation, each CWNA-coupled to its bias counterpart (`clock_rate_q`, `system_time_rate_offset_q`). Under BSSD both rate states are omitted, because same-system single differencing cancels them.

Receiver clock $c\,\mathrm{d}t_r$. Single state, in metres. Its process noise is large (`clock_bias_q` $\sim 10^4\,\mathrm{m}^2$ per second), so the state is in effect re-estimated each epoch. This follows oscillator drift without imposing a stochastic model that may not match the receiver's oscillator. Under the BSSD (between-satellite single-difference) parameterisation (Section 4.1) the receiver clock is unobservable and the state is not allocated; neither are the inter-system offsets below, so PPP-RTK SD mode has no receiver-side clock block.

Inter-system time offsets $c\,\delta t_r^{(g)}$. One per active constellation other than GPS, which is the fixed reference in `StateManager`. These absorb the per-constellation clock biases that arise from mixing analysis-centre clock products across systems ("GAL − GPS time offset," etc.).

Tropospheric ZWD and gradients. ZWD_r is the zenith wet delay. The zenith hydrostatic delay ZHD_r is deterministic, from the Saastamoinen model [3]. The total slant tropospheric delay is

$$T_r^i \;=\; m_h(\theta_r^i)\,\mathrm{ZHD}_r \;+\; m_w(\theta_r^i)\,\mathrm{ZWD}_r \;+\; m_w(\theta_r^i)\cot\theta_r^i\big[g_N\cos\alpha_r^i + g_E\sin\alpha_r^i\big], \tag{3}$$

with elevation θ_r^i, azimuth α_r^i, and mapping functions m_h, m_w: the Niell tables by default (labelled `Gmf` in the configuration), or VMF3 with GPT3 coefficients when configured. The gradients (g_N, g_E) are optional.

Per-satellite slant ionosphere $I_{r,1}^i$. PPP-RTK only (uncombined processing). One state per visible satellite, expressed at the L1 frequency; signal-specific delays follow Equation (1). `StateInitializer::computeIonosphereSeed` seeds it from the geometry-free combination of the satellite's BIA-covered code residuals. With fewer than two covered codes on different bands, the seed is zero and the initial standard deviation is inflated tenfold. The initial variance, like the process noise, is scaled by the obliquity factor. The process noise is a random walk whose zenith spectral density `ionosphere_q` is projected onto the slant by the squared obliquity factor $1/\sin^2\theta_r^i$ (the ionospheric process-noise model in the estimation document), so low-elevation states vary faster. These explicit per-satellite states let PPP-RTK keep integer ambiguity recovery without forming the ionosphere-free combination, which amplifies code noise by a factor of ~ 3. Under `estimation.enable_iono_rate` (default off), each visible satellite also carries a slant-ionosphere rate, so that the pair forms a constant-velocity block $[\,I_{r,1}^i,\ \dot{I}_{r,1}^i\,]$. The rate $\dot{I}_{r,1}^i$ has no direct observation ($H = 0$); only the F-coupling into the level state drives it. It has its own spectral density `process_noise.iono_rate_psd` and its own slip response (`reinflateIonosphereRateVariance`), and lets the filter track a linear slant ramp rather than a pure random walk. The rate state is a satellite-specific

dynamic row, not a core state: it is appended with its level state, pruned with it and treated with the other per-satellite states in gap recovery, but after prunes and re-adds it need not sit next to the level state, so it is addressed only through its own map (`iono_rate_indices_`).

Carrier-phase ambiguities $N^i_{r,s}$. One state per phase-tracked (i, s) pair, in metres (the state value is λN; the `nl_wl` basis below is the exception, carrying integers in cycles). The ambiguities are float through the prediction and update cycle; when AR is enabled, the LAMBDA-family algorithms in `filter/AmbiguityResolver` [5] resolve them to integers afterwards. A cycle slip calls `StateManager::resetAmbiguity`, which decorrelates the state rather than repairing it: the ambiguity's covariance row and column are zeroed, its diagonal is reset to σ_m^2, and the key is marked pending re-seed. The state value is left unchanged by the reset; `ensureRequiredStates` (in `MeasurementModel`) then overwrites it with a geometry-based seed. This is the project's reset-over-repair convention: a slipped ambiguity has no trustworthy correlation with the rest of the state, so its cross-covariances are cleared. A reset for a key that is not in the state (for example, a signal excluded by the band filter) is refused and creates no state.

Receiver code biases $B_{r,s}$. These absorb the receiver-side inter-frequency biases that the OSB convention does not reference to zero. They are not one per signal. `StateInitializer::resolveReceiverBias` keys them per (system, frequency band): both bands of the system's clock-reference signal pair map to one per-system differential code bias (`"<system>_DCB"`), each band with its own coefficient, and every other band gets its own inter-frequency bias (`"<system>_IFB<band>"`). Like the receiver clock they are receiver-common, so under BSSD they receive no process noise and the SD transform zeroes their columns. On the network path, under `network.second_code_channel` or `processing.channel_mode:all`, a code that is not the station's latched canonical code on its band gets a separate receiver inter-signal code bias (`"<system>_ISC_<signal>"`, unit coefficient, GLONASS excluded), stored in the same map (card SCC). On the network path these columns have no Kalman time update at all, because the producer runs no `KalmanFilter` (the filtering document); they are re-estimated each epoch unless `network.carry_rx_code_bias` is armed. That knob (loader default off; armed in the current golden configurations) adds a diagonal random-walk carry ($\mathrm{var} + q\,\Delta t$) held in the network run context rather than in `StateManager`. Two guards protect it: at load, the loader forces the carry off unless the code datum is time-linked (`network.code_datum_time_linked`); at inject time, an innovation gate (`rxCbInjectGate` in `filter/NetworkRunTypes`) withholds the fold when the carried level and the station's own estimate disagree at receiver-reset scale.

Per-slot GLONASS inter-frequency bias $B^{(g,b)}_{r,k}$. Optional, off by default. GLONASS FDMA assigns each slot $k \in [-7, +6]$ a distinct carrier frequency $f = f_{\mathrm{base}} + k\,\Delta f$, so the receiver group delay differs per slot and a single per-band IFB state cannot represent it. With `estimation.glonass_ifb.enable = true` and its prerequisite `estimation.estimate_ifb` (without which the loader warns and the flag is inert), `resolveReceiverBias` instead creates one state per (FDMA band $b \in \{\mathrm{G1}, \mathrm{G2}\}$, slot k) with unit measurement coefficient, keyed `"GLONASS_IFB<band>_k<±slot>"`. The central channel $k = 0$ is the held datum and has no state; the GLONASS inter-system time offset absorbs its constant. The CDMA bands (G3/G4/G6) have a slot-independent frequency; with the flag on, the GLONASS branch of `resolveReceiverBias` returns no receiver bias key for them (it handles every GLONASS signal and does not fall through to the per-band path). With the flag off, GLONASS carries no per-band IFB state on a non-reference band. Unlike the receiver-common biases, these columns survive between-satellite single differencing, because two GLONASS satellites

on different slots do not cancel. `BssdTransform` and `StateManager` therefore exempt them from the receiver-block zeroing and from the process-noise suppression (via `isGlonassPerSlotIfbKey`); only the inter-slot differences are estimable in SD.

Satellite code biases B_s^i. Per (satellite, signal) pair, but only for a pair whose OSB the BIA file does not supply, typically a third-frequency signal in a triple-frequency setup. This fallback rule is the user-path and network-V1 behaviour. In the network producer's satellite-OSB mode (`network.estimate_sat_code_bias`, which `network.estimate_iono_sh` forces on unless `network.external_sat_code_bias` is armed), the state covers every raw code (i, s) rather than only the uncovered pairs, and the external `.BIA` is not used, so the network's own estimate is the single code-bias source. The datum that makes it estimable is derived in the network-mode document. These states are weakly stochastic (`sat_bias_q` $\sim 10^{-10}$), because the underlying physical biases are nearly stationary over a day.

Satellite phase biases $\bar{b}_s^i$ (network mode). Present only in the network producer (`ProcessingMode::PPP_RTK_Network`) and gated by `estimation.estimate_sat_phase_biases`. The JSON default is off and the flag is inert in every other mode. In the one mode where the states exist, the network-producer loader forces the flag on unconditionally (the branch in `GnssConfigLoadNetwork` that logs *forcing estimate_sat_clocks + estimate_sat_phase_biases*), and `GnssConfig::applyNetworkConsumerCouplings` turns it off again for the closed-loop consumer, which consumes these states rather than estimating them. The user side imports the satellite phase bias as a known SSR correction (the SSR-corrections document) or absorbs it into the float ambiguity. The network instead carries an explicit per-(satellite, signal) phase-bias state on the undifferenced phase rows: the estimable bias of Odijk [2], which lumps the network pivot-receiver ambiguity. On a single station's row it is collinear with the per-receiver float ambiguity; only the joint network estimator separates it, across stations (the network-mode document). It has its own state map, distinct from the satellite code biases above (`StateManager::sat_phase_bias_indices_`), keyed `"<sat>|<signal>|PB"`. It is near-stationary between epochs, with rate `sat_phase_bias_q`; on the network path that rate inflates a shared-border carry prior rather than entering a Kalman $\boldsymbol{Q}(\Delta t)$ (see the satellite-clock paragraph below).

Receiver phase biases $\delta_{r,s}$ (network mode). The receiver-side counterpart of the satellite phase bias, present only in the network producer and gated by `network.phase_datum_eq34` (the Eq-34 / single-receiver-chain phase datum; see the network-mode chapter). With its own estimable state $\delta_{r,s}$, the receiver fractional phase bias no longer lumps into the emitted satellite phase bias, so the emitted between-satellite reference stays integer-consistent across all satellites (each chain hop contributes only an integer). It is per (receiver, system, signal), keyed `"<system>_RPB_<signal>"` in its own map (`StateManager::receiver_phase_bias_indices_`). Unlike every other bias block described here, these are local states, each observed by a single station. A knob sets whether they carry across epochs. At `network.phase_datum_eq34_rpb_q_m2s` $= 0$ each $\delta_{r,s}$ is re-estimated every epoch (the Card-3 behaviour). At the non-zero default ($3 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$) the producer folds a local time-linking prior toward the previous estimate with variance var $+ q\,\Delta t$. The prior suppresses the per-epoch bounce that a weakly observable local state otherwise injects, anti-correlated, into every individual float ambiguity. That carry lives outside `StateManager`, in the network run context (the `rpb_carry` map of `app/NetworkRunContext`): it is read during the parallel block assembly, harvested after the solve, and erased for any (i, s) whose chain frame stepped at that epoch's election (`filter/NetworkChainDatum`). Under `network.carry_invariant` the carry is off, and the loader

warns that the q is inert. In either case the receiver-side wander goes into $\delta_{r,s}$, and the time-linked integer information stays in the ambiguity carry.

Satellite clocks $c\,\Delta\delta t^i$ (network mode). Also network-producer-only, gated by `estimation.estimate_sat_clocks`. Like the phase bias above, it is off by JSON default and inert elsewhere, and the network-producer loader forces it on unconditionally at the same site. The network estimates an explicit per-satellite clock correction $\Delta\delta t^i$ (in metres, relative to the a-priori precise or real-time product clock already removed from `obs.modeled`), so the emitted clocks stay consistent with the chosen S-system datum. There is one state per satellite, with no signal field. The satellite orbit stays fixed from the product; only the clock is estimated. The state has its own map (`StateManager::sat_clock_indices_`), keyed `"<sat>|SC"`. Its cross-epoch rate `sat_clock_q` is near-white, but on the network path it is not a Kalman $\boldsymbol{Q}(\Delta t)$ entry: the producer runs no `KalmanFilter` (the filtering document), and the rate instead inflates the shared-border carry prior as $\mathrm{var} + q\,\Delta t$. The same holds for `sat_phase_bias_q` and `sat_code_bias_q`. All three can be set through their legacy `filter.process_noise.*` keys or through the network-scoped `network.process_noise` class table, whose per-class fallbacks equal the legacy defaults. Between full solves the split-cadence fast tier (`network.fast_clock`, default off and currently gate-failed) may refresh the emitted clock. That tier estimates a clock delta in its own least-squares tick and creates no state-vector row, so it does not appear here; see the network-mode document.

Inertial augmentation. With IMU integration enabled, the core state gains nine entries: a three-axis attitude error and the three-axis accelerometer and gyroscope biases. The strapdown mechanisation propagates them, rather than the random-walk process model. The inertial-aiding document treats them in full.

Baseline differential troposphere. In baseline mode the core is the position alone; double differencing cancels the receiver clocks and inter-system offsets, and on a short baseline the ionosphere and troposphere, so none of those states exists. An optional single differential zenith-wet-delay state $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}}r_{\mathrm{ref}}}$ follows the position (`baseline.estimate_differential_tropo`, Section 4.3). This is the only mode in which `StateManager` reads its `core_state_size` constructor argument; the PPP and PPP-RTK paths ignore it and derive the core layout from the configuration.

2.3 Dynamic indexing

`StateManager` addresses the variable-dimension blocks through key→index maps. `include/filter/StateKeySchema.hpp` defines every key format once, each with a `make` builder and, for the categories that are read back, a `parse` helper (the ionosphere and inter-system time-offset keys have no parser, since the key is the bare satellite or system name). Some call sites still build keys by string concatenation rather than through the builders (for example the receiver IFB and DCB keys in `resolveReceiverBias` and the double-difference keys in `MeasurementModel`, `DdPivotManager` and `KalmanFilter`); the strings they produce match the schema formats. The formats are:

- iono key: `"<sys><prn>"`, e.g. `"G05"` (delimiter-free);
- ambiguity key: `"<sys><prn>_<signal>"`, e.g. `"G05_L1C"`, split on the *first* underscore, so an iono-free combination `"G05_L1C_L5Q"` retains its compound signal;
- satellite signal-bias key (code): `"<sys><prn>|<signal>"`, e.g. `"G05|C1W"`, with a pipe delimiter and its own map, disjoint from the underscore ambiguity namespace;

- satellite phase-bias key (network): `"<sys><prn>|<signal>|PB"`, e.g. `"G05|L1C|PB"`, where the trailing `|PB` keeps it disjoint from the code-bias namespace, in its own map (`sat_phase_bias_indices_`);
- receiver phase-bias key (network): `"<system>_RPB_<signal>"`, e.g. `"Galileo_RPB_L5Q"`, where the `_RPB_` infix keeps it disjoint from the receiver code-bias namespace (which matches on `IFB`/`DCB`), in its own map (`receiver_phase_bias_indices_`);
- satellite clock key (network): `"<sat>|SC"`, e.g. `"G05|SC"`, per satellite rather than per signal; the trailing `|SC` discriminator gives the clock its own map (`sat_clock_indices_`);
- receiver code-bias key: `"<system>_IFB<band>"` or `"<system>_DCB"`; the per-slot GLONASS variant appends an FCN marker, `"GLONASS_IFB<band>_k<±slot>"` (e.g. `"GLONASS_IFB1_k-4"`);
- receiver inter-signal code-bias key (network, card SCC): `"<system>_ISC_<signal>"`, e.g. `"GPS_ISC_C1W"`, stored in the receiver code-bias map; the `_ISC_` infix matches neither the IFB nor the DCB parser;
- inter-system time-offset key: the bare system name (`"Galileo"`, ...; GPS is the reference and never appears);
- double-difference ambiguity key (baseline): `"DD|<cons>|<sat>|<code>|<pivot>"`.

Addressing states by key rather than by position removes the index bookkeeping errors of hand-managed dynamic state vectors, and lets the BSSD and DD S-transforms (Section 4) operate as block operations on the relevant key sets.

Two further categories in the schema header are absent from the list above, because they are not rows of $\boldsymbol{x}$. The first is the BSSD pivot absolute-seed lookup key (`"<cons>|<band>"`). Its builder has no production caller and it has no parser: `MeasurementModel` builds the `pivot_abs_seeds` keys by concatenation, with the full phase signal code in place of the band (`"GPS|L1C"`), plus narrow-lane and wide-lane variants under the `nl_wl` basis. The second is the shared spherical-harmonic VTEC coefficient key (`"SH|<n>|<m>|<C|S>"`, with the second shell on the distinct `SH2` prefix), which names columns of the network estimator's shared border rather than `StateManager` states and has a `make`/`parse` pair. Both appear in the generated key-format table of the appendix, which is derived from the schema header by a script and is therefore the completeness reference for the list above.

Ambiguity basis. By default the ambiguity block carries one per-band float ambiguity per tracked (satellite, signal), in metres. Setting `estimation.state_ambiguity_basis = "nl_wl"` switches the whole run to a narrow-lane / wide-lane parameterisation. Each satellite has a reference band, the lowest-numbered band it is observed on that epoch, and carries one narrow-lane state tied to that band (key `"<sat>_NL_<band>"`) and one wide-lane state for each other observed band, paired with the reference band (`"<sat>_WL_<a>_<b>"`, with the band digits in ascending order, semantically $N_a - N_b$). A change of reference band produces new keys, and the old states are pruned. Both kinds are in cycles rather than metres, so the wide-lane and LAMBDA arithmetic stays native to integers; the Jacobian carries the wavelength scaling. The two bases never coexist in a single state vector.

3 Observation equations

This section gives the full *undifferenced* observation model that the filter linearises around at each epoch. The PPP-RTK parameterisation is uncombined: per-signal pseudorange and carrier phase are consumed directly, without forming ionosphere-free or wide-lane combinations as a preprocessing step. Combinations (IF, MW, GF) are still computed for diagnostic and slip-detection purposes

(the cycle-slip document; see `slip/MwSlipDetector`, `slip/GraphicSlipDetector`) but they do not enter the measurement update path. This holds whenever the uncombined parameterisation is active; the loader forces `uncombined_enabled=true` for the whole PPP-RTK family (user, network producer, network consumer) in `GnssConfig::loadProcessingSection_`; the same site forces it for the baseline double-difference modes, which process each signal independently (Section 4.3). Plain PPP mode leaves the flag at its default `false` and instead builds ionosphere-free combined observables (keys such as `C1W_C5X`) that *do* feed `MeasurementModel::buildJacobianRow` and the update; the sequel is written for the uncombined case.

3.1 Code (pseudorange) observation

For receiver r tracking satellite i on signal s, the observed pseudorange is

$$\boxed{P^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;+\; \mu_s\, I^i_{r,1} \;+\; B_{r,s} - B^i_s \;+\; \varepsilon^P_{r,i,s}.} \tag{4}$$

3.2 Carrier-phase observation

The corresponding carrier-phase observation, scaled to metres by λ_s, is

$$\boxed{\Phi^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;-\; \mu_s\, I^i_{r,1} \;+\; \lambda_s\, N^i_{r,s} \;+\; b_{r,s} - b^i_s \;+\; \varepsilon^\Phi_{r,i,s}.} \tag{5}$$

The defining structural difference between Equations (4) and (5) is the sign of the ionospheric term: $+\mu_s I$ on code, $-\mu_s I$ on phase. Three additional remarks:

1. The phase ambiguity $N^i_{r,s}$ is integer in principle but is contaminated by the receiver-plus-satellite phase biases $b_{r,s} - b^i_s$. In a *float* solution these biases are absorbed into $N^i_{r,s}$, leaving a non-integer state with no structural advantage over the code observation. Recovering integer-valued ambiguities therefore requires either external Fractional Cycle Bias (FCB) products or differencing (Section 4), which cancels common-mode bias terms.
2. The clock product $c\,\mathrm{d}t^i$ is read from the analysis-centre CLK file (post-processing) or refined from the broadcast clock by a streamed SSR correction (real-time, the SSR-corrections document) and treated as deterministic in the filter; any residual product error appears on the observation and is absorbed by the receiver clock state.
3. The system time offset $c\,\delta t_r^{(g(i))}$ is zero by construction when $g(i)$ is the reference constellation (default GPS). Together with the receiver clock $c\,\mathrm{d}t_r$ this means a multi-constellation run carries $|\mathcal{G}|$ clock-like states in total.

3.3 Geometric range and linearisation

The geometric range $\rho^i_r = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|_2$ enters Equations (4) and (5) non-linearly through the receiver position $\boldsymbol{r}_r$. The filter linearises about the current state estimate $\hat{\boldsymbol{r}}_r$:

$$\rho^i_r \;=\; \hat{\rho}^i_r \;-\; (\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}(\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r) \;+\; \mathcal{O}\big(\|\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r\|^2\big)\,, \tag{6}$$

with the unit line-of-sight

$$\boldsymbol{e}^i_{\hat{r}} \;:=\; \frac{\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r}{\|\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r\|_2}. \tag{7}$$

The Jacobian row $\boldsymbol{H}_{(i,s)}$ contributing to the design matrix at the position columns is therefore $-(\boldsymbol{e}^i_{\hat{r}})^{\mathrm{T}}$. For sub-metre residual targets this single linearisation per epoch is sufficient; the engine

optionally relinearises within the epoch (`processing.relinearize_geometry`, which defaults to off in code and is enabled in the PPP, PPP-RTK, baseline, and real-time configs but left off in the six network-producer configs) to handle long initial position offsets.

The satellite position $\boldsymbol{r}^i$ is itself a function of the transmission time. Rather than iterate the emission time against the geometric range, which depends on $\boldsymbol{r}_r$, `Geometry::computeEmissionState` solves it in a single step from the *pseudorange*: the time of flight is $\tau = P/c$, subtracted (with the satellite clock) from the receiver epoch. Because the receiver clock bias $\mathrm{d}t_r$ enters the receiver epoch as $+\mathrm{d}t_r$ and the pseudorange as $+c\,\mathrm{d}t_r$, it cancels exactly, leaving a residual $(I+T+\varepsilon)/c \approx 30$–$60$ ns of satellite displacement, sub-millimetre and below any PPP threshold. This deliberately avoids a range-based fixed point: iterating against ρ_r^i would re-introduce an uncancelled $\mathrm{d}t_r$ and drive a positive-feedback loop through the filter's receiver-clock state. The only second pass (the `model_corrections.refine_emission_time` block in `Geometry::computeSatelliteGeometry`) re-runs `computeEmissionState` on a cleaned pseudorange, having first subtracted the slant hydrostatic delay and the IF-combined ATX projection from the seed, a pseudorange-cleanliness refinement rather than a range-versus-position iteration.

3.4 Stochastic model

Observation noise covariance is structured per epoch as a block diagonal matrix with entries

$$\sigma^2_{r,i,s} \;=\; \sigma^2_{0,\mathrm{type}(s)} \;\cdot\; w(\theta_r^i)^{-1} \;\cdot\; \xi(\mathrm{C/N_0 \text{ at } } (r,i,s)), \tag{8}$$

combining a per-type baseline ($\sigma_{0,\mathrm{code}}$, $\sigma_{0,\mathrm{phase}}$ from `filter.measurement_noise`), an elevation-dependent down-weighting factor $w(\theta_r^i)^{-1} = (\sin\theta_r^i)^{-2p_{\mathrm{type}}}$ whose exponent is *continuous* and *type-dependent*: code and Doppler rows take `filter.code_elevation_power`, carrier-phase rows `filter.phase_elevation_power`, with the weighting strategy itself selected by `filter.stochastic_model` (derived, together with the low-elevation guard that leaves σ_0^2 unamplified, in the stochastic-model section of the quality-control document), and an optional carrier-to-noise ratio scaling ξ parameterised by `snr_model_params`. Cross-correlations between observations are assumed zero in the undifferenced parameterisation; differencing introduces structured correlations handled in Section 4.

4 Differencing transformations

PPP-RTK and baseline modes do not consume the undifferenced observations of Equations (4) and (5) directly. Each forms a linear combination across satellites (BSSD), receivers (BRSD), or both (DD), absorbing common-mode terms into the cancellation and recovering integer-friendly residual structure. This section gives both the algebraic cancellations and the matrix-form S-transforms that map undifferenced parameters to differenced ones.

4.1 Between-satellite single difference (BSSD)

For each constellation g, choose a pivot satellite p_g (elected by `PivotManager::selectPivotCandidates` under the "sticky pivot" retention policy; the selection ordering is detailed in the *Pivot selection* note below). For any non-pivot satellite $i \in \mathcal{S}_g \setminus \{p_g\}$ and signal s, define

$$\begin{aligned} \Delta P^{i,p_g}_{r,s} &:= P^i_{r,s} - P^{p_g}_{r,s}, \\ \Delta \Phi^{i,p_g}_{r,s} &:= \Phi^i_{r,s} - \Phi^{p_g}_{r,s}. \end{aligned} \tag{9}$$

Substituting Equations (4) and (5) and cancelling the receiver-side common terms ($c\,\mathrm{d}t_r$, $c\,\delta t_r^{(g)}$, $B_{r,s}$, $b_{r,s}$) yields

$$\boxed{\begin{aligned}\Delta P_{r,s}^{i,p_g} &= \Delta\rho_r^{i,p_g} - c\,\Delta\mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} + \mu_s\,\Delta I_{r,1}^{i,p_g} - \Delta B_s^{i,p_g} + \Delta\varepsilon^P, \qquad (10)\\ \Delta\Phi_{r,s}^{i,p_g} &= \Delta\rho_r^{i,p_g} - c\,\Delta\mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} - \mu_s\,\Delta I_{r,1}^{i,p_g} + \lambda_s\,\Delta N_{r,s}^{i,p_g} - \Delta b_s^{i,p_g} + \Delta\varepsilon^\Phi. \qquad (11)\end{aligned}}$$

The cancellations are exact only when both observations enter the filter with the same modelling: same tropospheric mapping (which is why ΔT is small but not zero), same antenna PCV correction, and so on.

The receiver-bias cancellation of $B_{r,s}$ in Equation (10) holds for CDMA constellations, whose group delay is a single per-signal receiver term. GLONASS is the exception: its FDMA inter-frequency biases are per-FCN-slot states, so two satellites on different slots carry different receiver delays and the term does *not* cancel between them: a residual $B_{r,s}^{(k(i))} - B_{r,s}^{(k(p_g))}$ survives the single difference. Accordingly `BssdTransform` zeroes the receiver-DCB columns after differencing *except* the per-slot GLONASS IFB states (guarded by `StateKeys::isGlonassPerSlotIfbKey`), and `PivotManager::buildEpochMaps` drops GLONASS from the BSSD candidate maps altogether, not merely from pivot election, so with the flag off the constellation contributes no single-differenced row at all. GLONASS BSSD is default-OFF (`estimation.between_satellite_sd.enable_glonass`); when enabled, Equation (10) acquires the surviving inter-slot IFB difference as an additional estimated column.

Pivot selection. Election is not by elevation alone. The primary key is a *hard* lexicographic tier on phase-signal coverage: a full-coverage candidate outranks any partial-coverage candidate regardless of its score. Only within a coverage tier does the additive score $-\min\bigl(1, \sqrt{\sigma^2_{\hat{N},\max}}\bigr) + \mathrm{el}/90 - \frac{1}{2}\mathbb{1}[\mathrm{el} < 20^\circ]$ apply, in which float-ambiguity convergence $\sqrt{\sigma^2_{\hat{N},\max}}$ and elevation enter as co-equal terms (with a hard low-elevation guardrail below 20°). Layered on top is an ephemeris-freshness bar: a candidate whose broadcast $|\mathrm{obs} - \mathrm{toe}|$ exceeds its constellation's freshest qualified peer by more than `kPivotToeSpreadMaxS` (1200 s) is barred from *pivot* election (though still usable as a non-pivot), which prevents a stale-eph pivot from stepping every SD row it anchors. Candidates below `estimation.between_satellite_sd.pivot_min_elevation` are excluded outright. The baseline DD path scores differently: `DdPivotManager` ranks on $\mathrm{avg_snr} + 0.01 \cdot \mathrm{num_sigs}$ and does not use elevation.

Pivot transitions. A handover fires on two kinds of trigger. The visibility-driven one is simple: when the incumbent pivot leaves the visible set, p_g is re-elected from the remaining satellites. But the dominant one in practice is *event-driven*: the post-fit residual analyser calls `PivotManager::requestForcedHandover` when the pivot's *observation* slipped while its filter *state* is still intact (so the resting `isPivotDisturbed` check, which tests only that the pivot ambiguity sits at exactly $(0, 0)$, would not catch it). The forced request carries an exclusion set so the handover cannot bounce onto another offset satellite, a `retired_pivots_this_epoch_` backstop that terminates cascades, a same-epoch re-entry gate (`hasPendingForcedHandover`), and a re-inflation set for states that absorbed a coherent pivot-side code bias (the information-conserving transform below cannot evict that offset on its own). The baseline DD path exposes the symmetric `DdPivotManager::requestForcedRerank`. This event response is load-bearing precisely because the pivot's states are frozen at zero variance (below): a stiff, time-linked zero-q quantity requires a wired jump response.

On election the float ambiguities $\{\Delta N_{r,s}^{i,p_g^{\text{old}}}\}_{i,s}$ must be re-expressed relative to p_g^{new}. Where the new pivot already carries an established ambiguity in the group, the engine applies the linear S-transformation $\boldsymbol{x}^{\text{new}} = \boldsymbol{S}\,\boldsymbol{x}^{\text{old}}$ with $\boldsymbol{S}$ chosen so that $\Delta N_{r,s}^{p_g^{\text{new}},p_g^{\text{old}}}$ becomes the new reference (zero), preserving the integer information accumulated since the last fix; the covariance transforms as $\boldsymbol{P}^{\text{new}} = \boldsymbol{S}\,\boldsymbol{P}^{\text{old}}\,\boldsymbol{S}^{\mathrm{T}}$. When it cannot, because the new pivot has no existing ambiguity in the (cons, code) group (`DdPivotManager`, logged as "reset-and-reseed"), or a disturbed pivot has no healthy replacement in a single-satellite constellation (`PivotManager::handleDisturbedPivot` re-zeroes $\boldsymbol{x}$ and the corresponding rows/columns of $\boldsymbol{P}$ in place), the transform is skipped and the stale rows are discarded and re-seeded. In the uncombined PPP-RTK parameterisation the same $\boldsymbol{S}\,\boldsymbol{P}\,\boldsymbol{S}^{\mathrm{T}}$ re-reference is applied not only to the ambiguity block but to *three* state groups (`reReferenceAmbiguities`, `reReferenceIonosphere`, and `reReferenceSatBiases`), which is why `BssdTransform` zeroes the pivot's ionosphere and sat-bias $\boldsymbol{H}$-columns after differencing, and why $\Delta I_{r,1}^{i,p_g}$ in Equations (10) and (11) is itself a pivot-referenced SD quantity, not an absolute slant delay. The S-basis constraint is realised operationally by freezing the pivot's ambiguity, ionosphere, and sat-bias covariances to exactly zero (and suppressing their process noise), the very condition `isPivotDisturbed` tests against.

`PivotManager::getCleanHandovers` classifies each transition as lossless-invertible (*clean*) or discontinuous (disturbed/forced/re-seeded), so the RTS smoother can inverse-reference across a clean handover rather than force a segment boundary; the losslessness claim above holds only for the clean subset. These are two distinct code paths sharing the algebra above: in PPP-RTK the pivot re-election and the "sticky-pivot" retention policy live in `filter/PivotManager`, whereas the baseline DD path performs the analogous re-reference through the ambiguity-rename S-transform in `filter/DdPivotManager`, governed by the `baseline.enable_pivot_handover` flag.

4.2 Between-receiver single difference (BRSD)

In baseline modes there is at least one rover r_{rov} and exactly one fixed-coordinate reference r_{ref}. The BRSD operator forms

$$\nabla P^{i}_{r_{\text{rov}}r_{\text{ref}},s} \;:=\; P^{i}_{r_{\text{rov}},s} - P^{i}_{r_{\text{ref}},s}, \tag{12}$$

and analogously for $\nabla\Phi$. The cancellations produced by $\nabla\cdot$ are dual to those produced by $\Delta\cdot$: BRSD cancels *satellite*-side common terms ($c\,\mathrm{d}t^{i}$, B_s^i, b_s^i) but preserves receiver clock and biases of *both* stations. No state-space estimation of satellite biases is required when ∇ is applied, since they disappear before reaching the filter.

4.3 Double difference (DD)

The composition $\nabla\Delta\cdot = \Delta\big(\nabla\,\cdot\big)$ is order-independent up to noise correlation: $\Delta\big(\nabla\,\cdot\big) = \nabla\big(\Delta\,\cdot\big)$. Applying it to the carrier-phase model and collecting common-mode cancellations:

$$\nabla\Delta\Phi^{i,p_g}_{r_{\text{rov}}r_{\text{ref}},s} \;=\; \nabla\Delta\rho^{i,p_g}_{r_{\text{rov}}r_{\text{ref}}} + \nabla\Delta T^{i,p_g}_{r_{\text{rov}}r_{\text{ref}}} - \mu_s\,\nabla\Delta I^{i,p_g}_{r_{\text{rov}}r_{\text{ref}},1} + \lambda_s\,\nabla\Delta N^{i,p_g}_{r_{\text{rov}}r_{\text{ref}},s} + \nabla\Delta\varepsilon^{\Phi}. \tag{13}$$

What survives in Equation (13) is precisely the geometry term (linear in the rover position perturbation), the differential atmospheric delays, and the integer DD ambiguity: the structural form that makes carrier-phase RTK possible.

Short vs. long baselines. On short baselines ($\lesssim 10\,\mathrm{km}$) the differential atmospheric terms in Equation (13) are negligible at the millimetre level and need not be modelled. As baseline length grows, the differential ionosphere $\nabla\Delta I$ scales linearly with separation and saturates at typically 1 m

for separations of a few hundred kilometres, swamping the carrier wavelength. Two mitigations are available in `GnssProcessor`:

- `baseline.use_iono_free` forms the dual-frequency ionosphere-free combination at the DD level, eliminating the first-order ionospheric term at the cost of a non-integer ambiguity (AR is automatically excluded for IF rows) and $\sim 3\times$ amplified code noise;
- `baseline.estimate_differential_tropo` adds a single $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}} r_{\mathrm{ref}}}$ state with slow random-walk dynamics, capturing the residual differential wet delay that survives the BRSD on baselines with significant altitude difference.

4.4 S-transform structure

Both BSSD and DD admit a matrix-form representation as $\boldsymbol{x}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{x}_{\mathrm{undiff}}$ with $\boldsymbol{S}$ acting as a block design transform. For BSSD restricted to the ambiguity block of one constellation g on one signal s, with $|\mathcal{S}_g|$ tracked sats, $\boldsymbol{S}$ is a $(|\mathcal{S}_g|-1)\times|\mathcal{S}_g|$ matrix of the form

$$\boldsymbol{S}_{\mathrm{BSSD}} \;=\; \begin{bmatrix} 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots & & & \\ 0 & 0 & \cdots & 0 & -1 & 0 & \cdots & 1 \end{bmatrix}, \tag{14}$$

where the column of -1's sits at the index of the pivot p_g. The DD operator is the *composition* of the two single-difference operators applied in sequence, $\boldsymbol{S}_{\mathrm{DD}} = \boldsymbol{S}_{\mathrm{BSSD}}\,\boldsymbol{S}_{\mathrm{BRSD}}$, realised in code as `BrsdTransform::apply` followed by `BssdTransform::differenceBrsd` (`include/measurement/BssdTransform.hpp`). Only on a fully populated satellite $\times$ receiver observation grid does this composition coincide with the Kronecker product $\mathbf{D}_{\mathrm{sat}} \otimes \mathbf{D}_{\mathrm{rx}}$ of the between-satellite and between-receiver differencing matrices acting on the vectorised observations, a useful mental model but not what the non-rectangular real-data case computes. The transformed covariance is $\boldsymbol{P}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{P}_{\mathrm{undiff}}\,\boldsymbol{S}^{\mathrm{T}}$, a generally dense matrix even when $\boldsymbol{P}_{\mathrm{undiff}}$ is diagonal (because each differenced row borrows from the same pivot column). This dense correlation structure is the algebraic reason DD-based filters cannot be naively implemented as independent per-pair scalar updates; the correctly correlated joint update is what unlocks integer ambiguity recovery via the LAMBDA decorrelation. The general theory and a thorough catalogue of admissible S-transforms is given in Teunissen [4] and in the rank-defect treatment of Jonge [1].

Implementation notes. Two facts about how `GnssProcessor` realises the transforms above are worth stating explicitly. First, the chosen S-system is never materialised as an explicit minimum-constraint $\boldsymbol{S}$-basis matrix; it is realised *implicitly* through the pivot-differencing operator (`measurement/BssdTransform`), a faithful, S-basis-invariant realisation of the uncombined parameterisation of Odijk [2]. The three regimes (undifferenced, BSSD, and DD) are dispatched polymorphically by a `DifferencingMode` strategy in `measurement/MeasurementBuilder`, not by inline config branches. Second, integer recovery on a DD row requires that all four contributing observations share the same RINEX signal code. The row is marked `integer_capable` only when all three conjuncts hold: `!brsd_aliased` on *both* the i_i and pivot BRSD entries (rover and reference agree on each, i.e. no cross-attribute alias) *and* the rover uses the same code on i_i as on the pivot ($\texttt{rov_code}_i = \texttt{rov_code}_{\mathrm{piv}}$). The third conjunct is independent of the `brsd_aliased` flags: a row can be un-aliased on both entries yet still carry a non-integer DCB/IFB residual if the rover mixes codes across i_i and the pivot. Rows failing any conjunct are carried as float and excluded from ambiguity resolution.

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